

Proven C Book

Modern C, built on the proven library

draft — translation in progress

v0.1.0 · last updated 2026-08-05

rubidus

rubidus@gmail.com https://github.com/rubidus-api/proven_c_book

What this book is for — not a list of syntax, but an eye for reading **contracts**. It talks about the machine before it talks about C, and brings in syntax only when the story asks for it.

Who it is for — someone starting programming for the first time, and someone who knows the syntax but is stuck at “so what do I actually build with this, and how”. No prior knowledge is assumed, and it reads without a computer in front of you.

How it reads — no exercises, no “try it yourself”. It moves through question-and-answer in the running text; review is handled by deeper questions at the start of the next chapter.

What it covers — the fundamentals (bits, the memory ladder, pipelines, encodings, floating point), then the first program and a working toolchain, values and control flow, memory and pointers, data layout, precision and contracts, multiple files and the preprocessor, and finally the five bugs C has been shipping for fifty years — and the proven library as one answer to them. C23 is the default throughout.

The promise — every listing is compiled and run on every build, and the printed output is that run’s output (cross-checked with GCC and Clang).

Translation status

This English edition is being translated from the Korean original, which is complete (12 parts, 58 chapters, appendices A–E and an index, about 570 pages). The translation is **in progress**: this volume currently carries the front matter, and chapters are added as they are translated.

Until then, the complete text is the Korean edition:

- PDF — [dist/proven_c_book-v0.1.0.pdf](#)
- Web — [rubicus-api.github.io/proven_c_book/ko/](#)

Chapter-level progress is tracked in the repository ([TRANSLATION.md](#)).

Preface

This book is an introduction to **proven**, a C library I wrote — and, honestly, a piece of promotion for it.

But to argue that something is worth using, you first have to show why it is needed. The problems **proven** addresses live in the subtle corners of C: the limits of representation, the rules of conversion, the lifetime of memory, the regions the standard declines to define. To someone who has not met those corners, **proven** looks like an unnecessary contraption. So I decided to explain the problems first. That is why most of this book is plain standard C, and why **proven** does not appear in earnest until Part XI. If, after the earlier chapters, you conclude that you do not need **proven** — that is a good outcome too.

There is also, in these pages, some measure of affection for the language. Try explaining why a fifty-year-old language is still here, and what its design gave up in order to get what it got, and affection follows on its own. I have not hidden that part.

Every listing in this book is really compiled and really run. The printed output was not transcribed by hand: it is what the machine produced when the book was built, cross-checked with two compilers (gcc and clang).

One more disclosure. **This book was written with AI as an assisting tool.** What to cover, in what order, under which principles, and what to leave out — all of that I decided, and the draft was written to those instructions and then reviewed by me. Technical claims were checked against the standard and primary sources, and the examples are verified by machine on every build: whoever wrote it, **code that does not run does not go into this book.**

Errors will remain nonetheless. Those are mine, not the tool's. If you find one, please tell me.

Copyright and contact

The text of this book (prose and figures) is licensed under **CC BY-NC-SA 4.0**. You may share and adapt it with attribution; commercial use is not permitted, and adaptations must carry the same license.

The example code is licensed under the **MIT license**. Anything you learn here can go into your own programs without restriction.

Contact is by email. Corrections are best filed through the GitHub repository — they stay on the record and other readers can see them.

Only error reports are accepted as contributions. Wrong statements, listings that do not behave as printed, typos and stale information are all welcome. Manuscript contributions — new chapters or sections — are not.

rubidus@gmail.com

https://github.com/rubidus-api/proven_c_book